DAY 20 | PRE-READ

Saving Versions and Packing Boxes

A light read before Day 20. About 15 minutes, no code required.

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Why read this?

Tomorrow you learn tools every software team uses daily: Git to track changes, pipelines to check them automatically, and Docker to package an app so it runs the same everywhere. You also add security checks that stop unsafe code before it ships.

1. Git: save points for your work

Video games let you save progress so you can go back if something goes wrong. Git gives your code save points, called **commits**. Each one records what changed, who changed it and why.

2. Branches: parallel drafts

When several people edit one document at the same time, it gets messy. Instead, each person works on their own copy, then the changes are carefully combined. In Git these copies are **branches**, and the request to combine them is a **pull request**, where teammates review the work first.

> **Key idea:** the main branch should always work. New work happens on branches and is reviewed before joining main.

3. Security guards at the door NEW (v4)

A pipeline runs checks every time someone pushes code. Some checks are guards: one looks for risky code, one checks that the libraries you use have no known holes, and one looks for passwords left in files. If a guard finds a problem, the door stays shut until it is fixed. An AI reviewer such as OpenAI Codex can leave helpful comments, but a person still decides to open the door.

4. Docker: packing the whole kitchen

Imagine shipping a recipe to a friend abroad. Their oven and ingredients are different, so the dish turns out differently. Now imagine shipping a sealed box containing the dish, the ingredients and the oven. That is a **container**. It runs the same on your laptop, a test server or the cloud.

Docker word	Plain meaning
Dockerfile	The recipe: step-by-step instructions to build the box
Image	The sealed box, built from the recipe
Container	The box opened and running
Registry	A warehouse that stores boxes so others can use them

5. A smaller box is a safer box NEW (v4)

Every extra item in the box is something that could break or be attacked. Good teams pack only what the app needs, and the app inside does not run with the master key (the root user).

6. Think about these before class

- Have you ever lost work because a file was overwritten? How would save points have helped?
- Why is a password in a code file dangerous even if the repository is private?
- Why might software that works on your laptop fail on someone else's?

Before class: install Git and Docker Desktop (see shared/Laptop_Setup_Guide.pdf). Junbi (準備): preparation